

Modification & Restoration
is an Art....❤️🧐



DREAM DRIVE

Connecting Enthusiasts
with Services



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Information Systems Engineering

Digital Drive and Young Car Enthusiasts

25-26J-534

Introduction



Now a days everything rely on digitalization. When exploring the automotive industry, the whole world specially the up countries are using digitalization effectively. But social representations regarding automotive industry and classic car modification create unique marketing opportunities that remain highly untapped in Sri Lanka.

This is where “Digital Drive” comes into play through recommending local garages with their specialty and connect car enthusiasts and garage owners in effective way.

The Research Problem

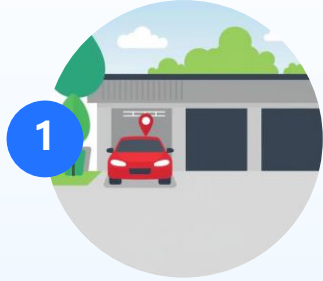
Car users in Sri Lanka still using traditional methods to find reliable garages such as time-consuming physical searches or rely on word-of-mouth recommendations.

Main Objective



There is a need to develop an intelligent, effective system that recommend with one click, connecting local garages and car enthusiasts.

Sub Objectives



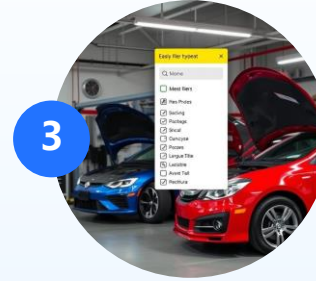
Instant garage matching

Find nearby garages within minutes based on your specific service needs.



Location-based search

Uses current location to connect with closest qualified service providers.



Specialized garages

Easily filter garages by car type and specific service requirements.



Rating system

Make informed choices through customer reviews and quality ratings.

The Business Model Canvas

Designed for:
Digital Drive and Young Car Enthusiasts

Designed by:
Dasuni K.S. – 25-26J-534

Date:
2025.12.30

Version:
V01

Key Partnerships



- Local garages and service centers
- Automotive parts suppliers
- Car clubs and enthusiasts communities
- Social media influencers in the automotive niche
- Payment gateway providers
- Location and mapping service providers.

Key Activities



- Platform development and maintenance.
- Geo-location integration for garage recommendations.
- Onboarding and verifying garage owners.
- Content moderation (videos, comments, reviews)
- Marketing and community engagement.

Key Resources



- Mobile and web application
- Geo-location and recommendation engine
- Garage dataset
- Content hosting infrastructure
- Marketing and community management team
- Technical development team

Value Propositions



- Simplifies finding specialized automotive services nearby.
- Connects car enthusiasts (age 18-30) with trusted local garages.
- Enables garages to showcase expertise through videos and profiles.
- Builds a social community for car enthusiasts to interact and learn.
- Inspire classic car restoration and modification culture among youth.

Customer Relationships



- Interactive platform with user profiles and engagement tools.
- Community-driven feedback and rating system.
- Personalized recommendations based on location and preferences.
- Social media integration for sharing and discussions.

Channels



- Website portal (primary platform)
- Mobile app
- Social media platforms (Instagram, TikTok, Facebook, YouTube)
- Automotive events and meetups.
- Partnerships with car clubs and garages.

Customer Segments



- Local garage owners and mechanics.
- Young car enthusiasts aged 18-30.
- Classic car collectors and restorers.
- Automotive content creators and influencers.

Cost Structure



- Platform development and maintenance.
- Marketing and user acquisition
- Server and hosting costs
- Customer support and moderation
- Partnership and licensing fees
- Content creation and promotional campaigns

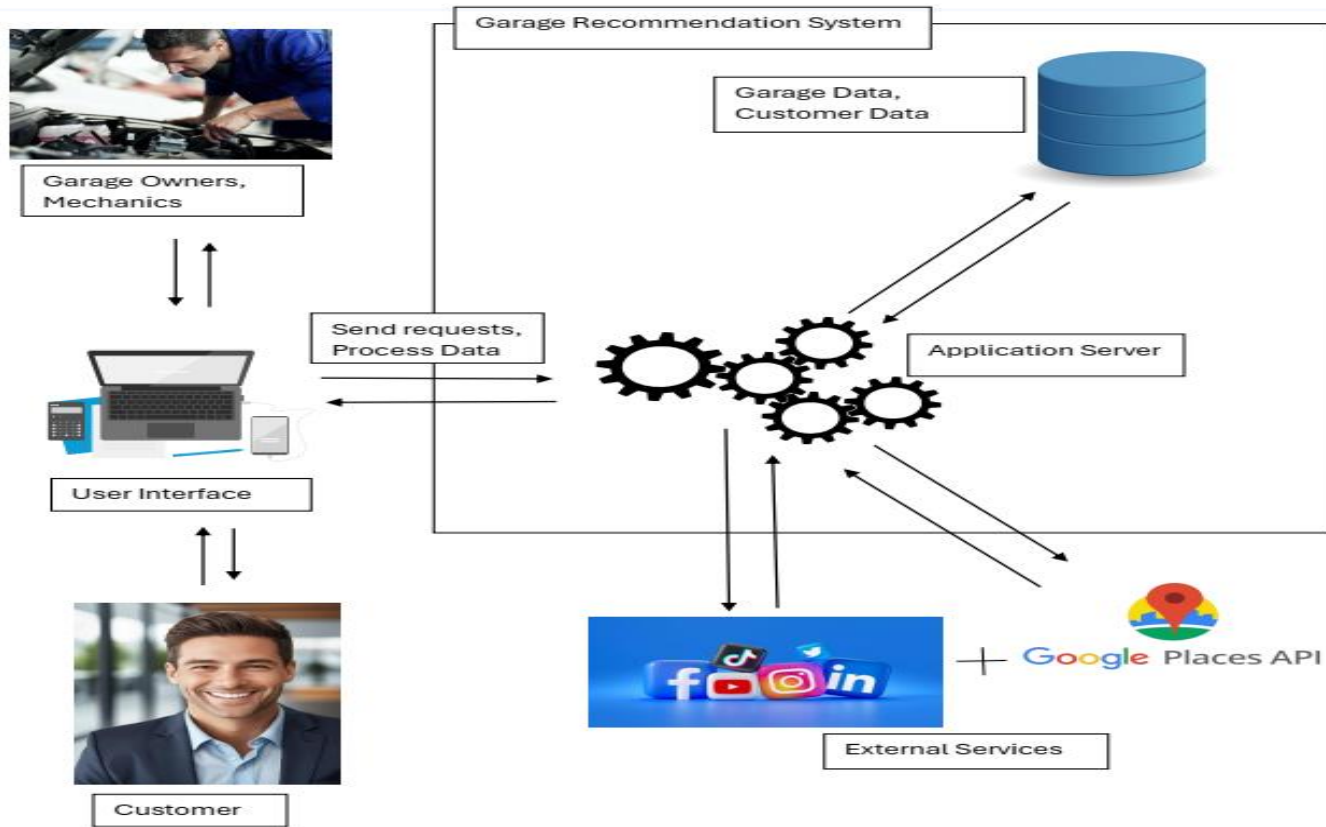
Revenue Streams



- Subscription fees from garages for premium listings.
- Advertising and sponsored content
- Commission on service bookings
- Affiliate marketing with part suppliers
- Event sponsorships and partnerships.

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Overall System Diagram



Research Novelty Component

Create a recommendation system to connect local garages with their specialty and car enthusiasts using machine learning for profiling and matching.

Research Focus & Gap Identification

Digital engagement gap

Limited online presence

Local garages lack digital visibility, making it difficult for enthusiasts to discover specialized automotive services.

Untapped market potential

Young car enthusiasts represent significant marketing opportunity that remains underserved in Sri Lankan automotive industry.

Disconnected communities

Car enthusiasts struggle to connect with qualified local service providers for specialized specialized vehicle needs.

Trust barriers

Absence of reliable review systems prevents quality assessment of garage services for potential customers.



Young car enthusiasts

Age 18-35

Primary target demographic with growing interest in automotive customization and digital solutions.



Digital Engagement

Connecting through social media platforms and interactive web application features.

Untapped Market

Significant opportunity in Sri Lankan automotive sector for specialized digital marketing and services.



Enthusiast Preferences

Focus on classic cars, sports modifications, and performance customization services.

Untapped marketing opportunities



1

Digital engagement gap

Young enthusiasts disconnected from local automotive services despite high interest in car culture.

2

Specialized services

Restoration and modification services lack visibility despite growing demand among 18-35 demographic.

3

Trust building

Ratings and reviews system creates transparency in transparency in previously word-of-mouth dominated industry.

4

Market expansion

Connecting enthusiasts with specialized garages creates new revenue streams for local businesses.

Research Problem

How can we create a recommendation system to connect local garages with their specialty and car enthusiasts using machine learning for profiling and matching?

Specific Objectives

Increase customer base – garage owners



Wider Reach

Connect with more potential customers through digital platform visibility.



Targeted Marketing

Attract specific customer segments based on service preferences and location.



Enhanced Visibility

Help garages showcase specialties to relevant audiences seeking specific automotive services.



Customer Retention

Build loyalty through ratings, reviews and consistent service quality tracking.

Create automotive marketplace – car users & garage owners

Connect Buyers and Sellers

Facilitate transactions between car enthusiasts and garage owners through integrated marketplace features.

Parts Exchange Platform

Enable trading of automotive parts, accessories, and specialized equipment within the Dream Drive community.

Allow customers to post projects while garages compete with quotes and portfolios. portfolios.

Service Bidding System

Feature completed classic and sports car projects to attract enthusiasts and potential potential clients.

Restoration Showcase

Methodology

Proposed System

Dream Drive Application

Web platform connecting car enthusiasts with local garages through one-click service matching.

Service Categories

Specialized garage listings for performance, painting, engine repairs, interior design, maintenance, wheel alignment, car care and accident repairs.

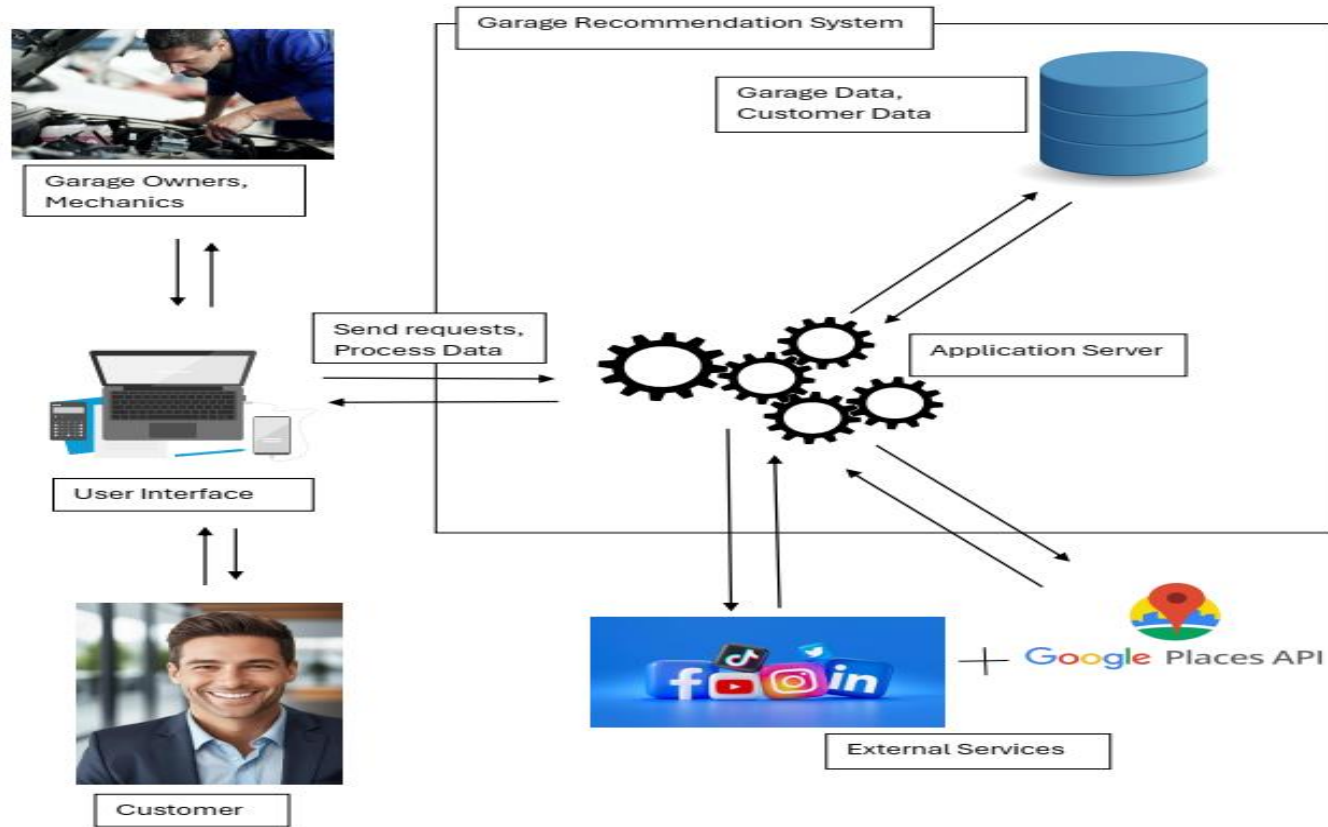
Quick Recommendations

Matches users with nearby garages within 1-2 minutes based on location and requirements.

Target Audience

Young car enthusiasts aged 18-35 seeking specialized automotive services.

System Diagram



Data Collection Techniques

01



Online Surveys and Questionnaires

Surveys will be distributed to young car enthusiasts through google forums and social media groups to collect data on preferences and behaviors.

02



Social Media Analytics

Analysis of engagement metrics across platforms to understand how young consumers interact with automotive content and identify trending topics and influencers.

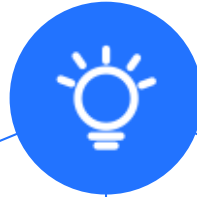
03



Focus Group Discussions

Structured conversations with selected participants to obtain qualitative insights into motivations, perceptions, and decision-making processes regarding automotive interests.

Tools and Techniques



Hardware & Support

- High performance Laptop/ PC
- Internet Connection
- Code Editor (VS Code)

Software Tools

- Python
- Anaconda
- Jupyter Notebook
- SQLite
- VS Code

Data Collection Tools

- Google Forms
- Social Media Platforms
- Google Places APIs

System Requirements Analysis

User Interface Requirements

Interactive dashboard for young car enthusiasts featuring customizable views, views, mobile responsiveness, responsiveness, and social media integration to enhance digital engagement with automotive content.

System Requirements

High performance Laptop/ PC,
Internet Connection,
Code Editor (VS Code),
and relevant software tools



Functional Requirements

User authentication, Project creation and showcase that the local workshops can upload multi-stage photo/video galleries of projects, User Interaction which is enthusiasts can 'Like' and 'Comment' on project showcases

Non- Functional Requirements

Performance: The platform must must load core content within 2 seconds, Security, Usability, Scalability

Current Progress – 80%

- Identified and collected the dataset.
- Analyzed the data according to the project requirements.
- Develop the advanced coding structure of the system.
- Developed the **machine learning python code for garage recommendation by fetching the customer geo-location**.
- Developed the **advanced coding structure for garage owners to create the profile of garage (register the garage), to upload images/ videos of their work showcase, to reply customers' questions and comments**.
- Developed the **advanced coding structure for customers to search registered garages, to view garage profiles, to ask questions and allow commenting, to view the location of the garage**.
- Developed the **advanced code to share, see distance, social media profile mentions and verification of recommended garages**.

Project Evidence

```
Research > GarageRecommendationSystem > newdashboard.py > ...
1 import streamlit as st
2 import sqlite3
3 import pandas as pd
4 import folium
5 from streamlit_folium import st_folium
6 import os
7
8 UPLOAD_DIR = "garage_media"
9 if not os.path.exists(UPLOAD_DIR):
10     os.makedirs(UPLOAD_DIR)
11
12 def init_db():
13     conn = sqlite3.connect('digital_drive.db', check_same_thread=False)
14     c = conn.cursor()
15     c.execute("""CREATE TABLE IF NOT EXISTS garages (
16         id INTEGER PRIMARY KEY AUTOINCREMENT, name TEXT, specialty TEXT,
17         lat REAL, lon REAL, phone TEXT)""")
18
19     c.execute("""CREATE TABLE IF NOT EXISTS garage_media (
20         id INTEGER PRIMARY KEY AUTOINCREMENT, garage_id INTEGER,
21         file_path TEXT, media_type TEXT)""")
22
23     c.execute("""CREATE TABLE IF NOT EXISTS comments (
24         id INTEGER PRIMARY KEY AUTOINCREMENT,
25         garage_id INTEGER,
26         username TEXT,
27         message TEXT,
28         parent_id INTEGER DEFAULT NULL)""")
29
30     conn.commit()
31     return conn
```

Code of Garage Profile Registration

```
Research > GarageRecommendationSystem > newdashboard.py > ...
47 st.subheader("📍 Pin the garage location")
48 center_lat, center_lon = 6.9271, 79.8612
49 m = folium.Map(location=[center_lat, center_lon], zoom_start=12)
50 m.add_child(folium.LatLngPopup())
51 map_output = st_folium(m, height=300, width=700, key="reg_map")
52
53 clicked_coors = None
54 if map_output.get("last_clicked"):
55     clicked_coors = (map_output["last_clicked"]["lat"], map_output["last_clicked"]["lng"])
56     st.success(f"Location Captured!")
57
58 with st.form("reg_form"):
59     name = st.text_input("Garage Name")
60     spec = st.selectbox("Specialty", ["Classic Restoration", "Modern Mods", "Performance Tuning"])
61     phone = st.text_input("Phone Number")
62     if st.form_submit_button("Create Profile"):
63         if clicked_coors and name:
64             c = conn.cursor()
65             c.execute("INSERT INTO garages (name, specialty, lat, lon, phone) VALUES (?, ?, ?, ?, ?)",
66                     (name, spec, clicked_coors[0], clicked_coors[1], phone))
67             conn.commit()
68             st.success("Your Profile Registered!")
69         else: st.error("Please click map and enter name.")
70
71 with tab_upload:
72     st.subheader("📁 Portfolio Uploads")
73     df_garages = pd.read_sql_query("SELECT id, name FROM garages", conn)
74     if not df_garages.empty:
75         selected_g = st.selectbox("Select your Garage", df_garages['name'])
76         g_id = int(df_garages[df_garages['name'] == selected_g]['id'].values[0])
77
78         uploaded_files = st.file_uploader("Upload Images or Videos",
79                                           type=['png', 'jpg', 'jpeg', 'mp4', 'mov'])
```

Send and retrieve data form
SQLite DB

Project Evidence

```
Research > GarageRecommendationSystem > newdashboard.py > ...
102
103 elif app_mode == "Garage Profiles":
104     st.title("🏠 Garage Profiles")
105     df_garages = pd.read_sql_query("SELECT * FROM garages", conn)
106
107     if not df_garages.empty:
108         selected_name = st.selectbox("🔍 Search for a Specialized Garage", df_garages['name'])
109         g_info = df_garages[df_garages['name'] == selected_name].iloc[0]
110         g_id = int(g_info['id'])
111
112         st.markdown(f"## 📁 {g_info['name']}")
113
114         # --- Media Gallery ---
115         st.subheader("🖼️ Portfolio & Work Showcase")
116         media_df = pd.read_sql_query(f"SELECT * FROM garage_media WHERE garage_id = {g_id}", conn)
117
118         if not media_df.empty:
119             # Separate Images and Videos for better layout
120             imgs = media_df[media_df['media_type'] == 'image']
121             vids = media_df[media_df['media_type'] == 'video']
122
123             if not imgs.empty:
124                 st.write("#### Photo Gallery")
125                 st.image(list(imgs['file_path']), width=220)
126
127             if not vids.empty:
128                 st.write("#### Video Gallery")
129                 for v_path in vids['file_path']:
130                     st.video(v_path)
131         else:
132             st.warning("No media uploaded for this garage yet.")
```

Code of garage portfolio media uploads

Code of Q & A part

```
st.subheader("💬 Community & Q&A")
tab1, tab2 = st.tabs(["Read & Reply", "Ask a New Question"])

with tab1:
    if not questions_df.empty:
        for _, q_row in questions_df.iterrows():
            # Display the Main Question
            with st.chat_message("user"):
                st.write(f"***{q_row['username']}** (Question)")
                st.write(q_row['message'])

            # 2. Fetch and display REPLIES for this specific question
            replies_df = pd.read_sql_query(
                f"SELECT * FROM comments WHERE parent_id = {q_row['id']}", conn
            )

            for _, r_row in replies_df.iterrows():
                # Display Replies indented or with a different color/icon
                st.markdown(f"""
                <div style="margin-left: 40px; border-left: 2px solid #ff4b4b; padding-left: 15px; margin-bottom: 10px;">
                <strong>👤 {r_row['username']} (Reply):</strong><br>{r_row['message']}
                </div>
                """, unsafe_allow_html=True)

            # 3. Add an Expander for the Reply Form
            with st.expander(f"Reply to {q_row['username']}"):
                with st.form(key=f"reply_form_{q_row['id']}"):
                    r_name = st.text_input("Your Name", value="Garage Owner")
                    r_msg = st.text_area("Your Reply")
                    if st.form_submit_button("Post Reply"):
                        if r_msg:
                            c = conn.cursor()
```

Project Evidence

```
RecommenderSystem.ipynb X Admin.py dashboard.py newdashboard.py garage_dashboard.py digital_drive.db
Research > GarageRecommendationSystem > RecommenderSystem.ipynb > import numpy as np
Generate + Code + Markdown | Run All | Restart | Clear All Outputs | Jupyter Variables | Outline
base Python 3.8

In [3]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import os

Out[3]: 0/0

In [4]: garages = pd.read_csv('dataset.csv')

Out[4]: 0/0

In [5]: garages.head(10)

Out[5]:
```

	name	phone	address	rating	reviews	place_id	lat	lng	category
0	AM Express Car Wash	0117 057 706	B263, Malabe, Sri Lanka	4.1	[[{"author_name": "Duleep Tharaka", "author_url": "https://www.google.com/profilecard/DuleepTharaka"}, {"author_name": "Chithra Jayasinghe", "author_url": "https://www.google.com/profilecard/ChithraJayasinghe"}]]	ChithraJayasinghe	6.906819	79.963148	car wash
1	Sabari Wash and Vacuum	077 314 4189	VVOW+445, Nugegoda 10100, Sri Lanka	4.5	[[{"author_name": "Rajin Rajasinghe", "author_url": "https://www.google.com/profilecard/RajinRajasinghe"}, {"author_name": "Chithra Jayasinghe", "author_url": "https://www.google.com/profilecard/ChithraJayasinghe"}]]	ChithraJayasinghe	6.870266	79.853363	car wash
2	Auto Shine Car Wash	077 773 8744	VW9+7W2, Colombo 00500, Sri Lanka	4.0	[[{"author_name": "Shephani Awarat", "author_url": "https://www.google.com/profilecard/ShephaniAwarat"}, {"author_name": "Chithra Jayasinghe", "author_url": "https://www.google.com/profilecard/ChithraJayasinghe"}]]	ChithraJayasinghe	6.880631	79.869777	car wash
3	Mr. Wax Car Wash	077 753 3955	Pannipitiya 10230, Sri Lanka	4.1	[[{"author_name": "Chathura Jayasinghe", "author_url": "https://www.google.com/profilecard/ChathuraJayasinghe"}, {"author_name": "Chithra Jayasinghe", "author_url": "https://www.google.com/profilecard/ChithraJayasinghe"}]]	ChithraJayasinghe	6.846650	79.952864	car wash

Python code for recommendation
(machine learning)

Testing Recommendation Results

```
Research > GarageRecommendationSystem > RecommenderSystem.ipynb > import numpy as np
Generate + Code + Markdown | Run All | Restart | Clear All Outputs | Jupyter Variables | Outline
base Python 3.8

def recommend(garage):
    matching_indices = garages_new[garages_new['tags'].str.contains(garage, case=False, na=False)].index

    if len(matching_indices) == 0:
        print(f"No garage found with tags containing '{garage}'.")
        return

    index = matching_indices[0]
    distances = sorted(list(enumerate(similarity[index])), reverse=True, key=lambda x:x[1])
    print(f"Recommended garages similar to '{garages_new.iloc[index]['name']}'")
    for i in distances[1:11]:
        garage_name = garages_new.iloc[i][0]['name']
        garage_rating = garages_new.iloc[i][0]['rating']
        garage_address = garages_new.iloc[i][0]['address']
        print(f"{garage_name} - Rate: {garage_rating}, Address: {garage_address}")

In [64]: recommend('car wash')

Out[64]: 0/0

In [65]: recommend('car wash')

Out[65]: 0/0

... Recommended garages similar to 'AM Express Car Wash':
Auto 52 - Battaramulla - Rate: 4.9, Address: 52 Main St, Battaramulla 10120, Sri Lanka
Platinum Car Care - Rate: 4.1, Address: 19B Kalalgoda Rd, Pannipitiya 10230, Sri Lanka
DF Autowash - Rate: 4.8, Address: Nawala Rd, Nugegoda 00500, Sri Lanka
Thilini Car Wash ( Quick Car Wash ) - Rate: 4.2, Address: 4 Narahenpita - Nawala Rd, Sri Jayawardanepura Kotte 10100, Sri Lanka
Bay Wash - Rate: 4.0, Address: 27 Colombo Plain road, Colombo 00400, Sri Lanka
Sabari Wash and Vacuum - Rate: 4.5, Address: VVOW+445, Nugegoda 10100, Sri Lanka
```

Project Evidence

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  JUPYTER

PS C:\Research> sqlite3 digital_drive.db
SQLite version 3.51.1 2025-11-28 17:28:25
Enter ".help" for usage hints.
sqlite> .tables
comments      garage_media  garages
sqlite> SELECT * FROM garages;
1|Lalith Auto Works|Classic Restoration|6.83923354129779|79.9033498764038|0771068960
sqlite> SELECT * FROM garage_media;
1|1|garage_media\bugatti-centodieci.jpg|image
2|1|garage_media\WhatsApp Video 2026-01-04 at 3.21.42 PM.mp4|video
sqlite>
sqlite> SELECT * FROM comments;
1|1|User1|Are you restoring BJ40s?|
2|1|Garage Owner1|Yes, we are restoring all types of classic cars.|1
3|1|Enthusiast_01|Are doing modifications for Glanza?|
4|1|Garage Owner1|Yes. We have uploaded a video on our profile...please check|3
sqlite>
sqlite>
sqlite> 
```

SQLite Database (Back-end)

Project Evidence

 **Garage Owner Dashboard**

[Register a Garage](#) [Upload Media \(Images/Videos\)](#)

 **Pin the garage location**



Map showing the location of the garage in Colombo, Sri Lanka. The map includes labels for various areas like Mattakkuliya, Modera, Aluthmawatha, Lunupokuna, Bloemendhal, Kelaniya, Maha Butthagamulla, Angoda, Ambatale, Kad, Kottu, Kotte, Sri Jayawardenepura, Arangala Junction, Malabe, Battaramulla, Koswatte, Thalahena, Mulleriyawa, Kalgahawatte, Udumulla, Kalapalawawa, Rajagiriya, Thimbirigasyaya, Narahenpita, Havelock Town, and Kuru. The map is credited to Leaflet and OpenStreetMap contributors.

Garage Name

Specialty

Classic Restoration

Phone Number

Create Profile

 **Garage Owner Dashboard**

[Register a Garage](#) [Upload Media \(Images/Videos\)](#)

 **Portfolio Uploads**

Select your Garage

Lalith Auto Works

Upload Images or Videos

 Drag and drop files here

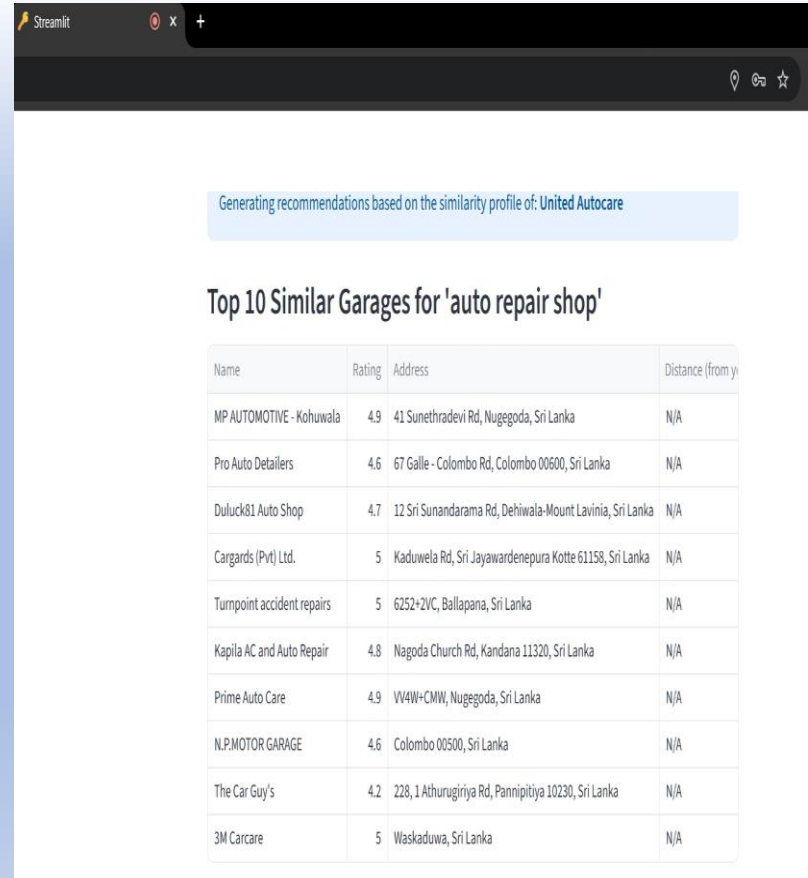
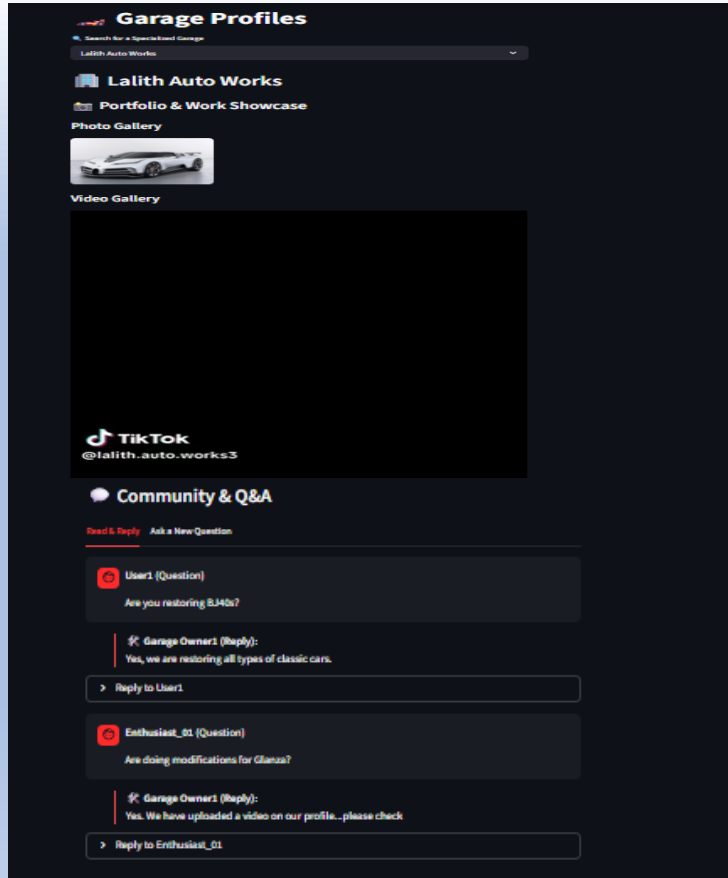
Limit 200MB per file • PNG, JPG, JPEG, MP4, MOV, MPEG4

Browse files

Save Media to Profile

UI/UX parts (Front-end)

Project Evidence



Future Work to be Completed

- Develop the advanced coding structure of the entire system.
- Develop the interface specially for young car enthusiasts regarding classic car modification and restoration by connecting social media platforms.
- Develop the UI/UX acceptable for an enterprise level product.

Thank You !

